

CHAPTER 1

NUMBER SYSTEMS

Rational Number:

A number which can be written in the form of $\frac{p}{q}$ (where $p, q \in \mathbb{Q}$ and $q \neq 0$) and $\frac{p}{q}$ is in

its lowest form. $\mathbb{Q} = \left\{ x \mid x = \frac{p}{q}, p, q \in \mathbb{Q} \wedge q \neq 0 \right\}$.

For example. $\frac{3}{5}, \frac{11}{19}, \frac{21}{2}$ are rational numbers.

Irrational Number:

A number which cannot be written in the form of $\frac{p}{q}$, where $p, q \in \mathbb{Q}$ and $q \neq 0$.

$\mathbb{Q}' = \left\{ x \mid x \neq \frac{p}{q}, p, q \in \mathbb{Q} \wedge q \neq 0 \right\}$.

For example. $\sqrt{2}, \sqrt{3}, \frac{3\sqrt{3}}{2}, \sqrt{\frac{17}{11}}$ are irrational

numbers.

SHORT QUESTION

What are terminating and recurring decimal called?

Terminating Decimals:

A decimal in which there are finite number of digits in its decimal part is called a terminating decimal.

For example. 1.23, 0.2, 7.95 are terminating decimals.

Recurring Decimals:

A decimal in which one or more digits repeat indefinitely.

For example:

$\frac{1}{3} = 0.333..., 7.2323..., 1.325325...$ are recurring decimals

Non-Terminating, Non-Recurring Decimal:

A decimal which neither terminates nor it is recurring.

For example: 0.1257..., 3.391...

3.141628732... is an important irrational number, which is the value of π (Pi). It denotes the constant ratio of the circumference of any circle to the length of its diameter.

$$\pi = \frac{\text{Circumference of any circle}}{\text{length of its diameter}}.$$

NOTE:

- (i) Every recurring decimal represents a rational number.
- (ii) An approximate value of π is $\frac{22}{7}$ and a better approximation is $\frac{355}{113}$.
- (iii) Every non-terminating, Non-recurring decimal represents an irrational number.

Real Number:

The union of rational and irrational numbers form real numbers. $\mathbb{R} = \mathbb{Q} \cup \mathbb{I}$,

PROPERTIES OF REAL NUMBERS**Binary Operation:**

A binary operation in a set A is a rule usually denoted by $*$ that assigns to any pair of elements of A , taken in a definite order, form another element of A .

Now we are discussing two important binary operations addition (+) and multiplication ($\cdot$) defined in the set of real numbers.

Addition Laws:		Multiplication Laws:	
(i) Closure Law: $\forall a, b \in \mathbb{R}$ such that $a + b \in \mathbb{R}$		(i) Closure Law: $\forall a, b \in \mathbb{R}$ such that $a \cdot b \in \mathbb{R}$	
(ii) Associative Law: $\forall a, b, c \in \mathbb{R}$ such that $a + (b + c) = (a + b) + c$		(ii) Associative Law: $\forall a, b, c \in \mathbb{R}$ such that $a(bc) = (ab)c$	
(iii) Additive identity: $\forall a \in \mathbb{R} \exists 0 \in \mathbb{R}$ such that $a + 0 = 0 + a = a$ Note: Zero is called additive identity of real numbers.		(iii) Multiplicative identity: $\forall a \in \mathbb{R} \exists 1 \in \mathbb{R}$ such that $a \cdot 1 = 1 \cdot a = a$ Note: “1” is called the multiplicative identity of real numbers.	
(iv) Additive inverse: $\forall a \in \mathbb{R} \exists -a \in \mathbb{R}$ such that $a + (-a) = (-a) + a = 0 \in \mathbb{R}$ Note: For additive inverse just change sign.		(iv) Multiplicative inverse: $\forall a (\neq 0) \in \mathbb{R} \exists \frac{1}{a} \in \mathbb{R}$ such that $a \cdot \frac{1}{a} = \frac{1}{a} \cdot a = 1 \in \mathbb{R}$ Note: For multiplicative inverse just take reciprocal.	
(v) Commutative Law: $\forall a, b \in \mathbb{R}$ $a + b = b + a$		(v) Commutative Law: $\forall a, b \in \mathbb{R}$ $a \cdot b = b \cdot a$	
<u>Distributive property of multiplication over addition:</u> $\forall a, b, c \in \mathbb{R}$ (i) $a(b + c) = ab + ac$ Left distributive law (ii) $(a + b)c = ac + bc$ Right distributive law			

Note: If any set possesses the above all properties then that set is called field. In sets $\mathbb{Q}$ (Rational Numbers), $\mathbb{R}$ (Real Numbers) and $\mathbb{C}$ (Complex Numbers) are known as fields.

PROPERTIES OF EQUALITY:	PROPERTIES OF INEQUALITIES: (order properties)
Equality of numbers denotes by "=" possesses the following properties.	(i) Trichotomy Property: $\forall a, b \in \mathbb{R}$ either $a = b$ or $a > b$ or $a < b$
(i) Reflexive property: $\forall a \in \mathbb{R}$	

$a = a$	
(ii) <u>Symmetric property:</u> $\forall a, b \in \square$ $a = b \Rightarrow b = a$	(ii) <u>Transitive Property:</u> $\forall a, b, c \in \square$ $a > b \wedge b > c \Rightarrow a > c$ $a < b \wedge b < c \Rightarrow a < c$
(iii) <u>Transitive property:</u> $\forall a, b, c \in \square$ $a = b \wedge b = c \Rightarrow a = c$	(iii) <u>Additive Property:</u> $\forall a, b, c, d \in \square$ (a) (i) $a > b \Rightarrow a + c > b + c$ (ii) $a < b \Rightarrow a + c < b + c$ (b) (i) $a > b \wedge c > d \Rightarrow a + c > b + d$ (ii) $a < b \wedge c < d \Rightarrow a + c < b + d$
(iv) <u>Additive Property:</u> $\forall a, b, c \in \square$ $a = b \Rightarrow a + c = b + c$	
(v) <u>Multiplicative Property:</u> $\forall a, b, c \in \square$ $a = b \Rightarrow ac = bc \wedge ca = cb$	(iv) <u>Multiplicative Property:</u> (a) $\forall a, b, c \in \square$ and $c > 0$ (i) $a > b \Rightarrow ac > bc$ (ii) $a < b \Rightarrow ac < bc$ (b) $\forall a, b, c \in \square$ and $c < 0$ (i) $a > b \Rightarrow ac < bc$ (ii) $a < b \Rightarrow ac > bc$ (c) $\forall a, b, c, d \in \square$ and a, b, c, d are all positive (i) $a > b \wedge c > d \Rightarrow ac > bd$ (ii) $a < b \wedge c < d \Rightarrow ac < bd$
(vi) <u>Cancellation Property w.r.t Addition:</u> $\forall a, b, c \in \square$ $a + c = b + c \Rightarrow a = b$	
(vii) <u>Cancellation Property w.r.t Multiplication:</u> $\forall a, b, c \in \square$ $ac = bc \Rightarrow a = b, c \neq 0$	

EXERCISE 1.1

Q.1 Which of the following sets have closure property w.r.t addition and multiplications?

(i) $\{0\}$

Solution:

Let $G = \{0\}$

As $0 + 0 = 0 \in G$

Hence, G possess closure property w.r.t addition

As $0 \times 0 = 0 \in G$

Hence, G possess closure property w.r.t multiplications.

(ii) $\{1\}$

FSD 2023

Solution:

Let $G = \{1\}$

As $1 + 1 = 2 \notin G$

Hence, G does not possess closure property w.r.t addition

As $1 \times 1 = 1 \in G$

Hence, G possess closure property w.r.t multiplication.

(iii) $\{0, -1\}$

Solution:

Let $G = \{0, -1\}$

$0 + 0 = 0$

$0 + (-1) = (-1) + 0 = -1$

$-1 + (-1) = -2 \notin G$

Hence, G does not possess a closure property w.r.t addition because all the sums do not belong to the set G

$0 \times 0 = 0$

$0 \times (-1) = (-1) \times 0 = 0$

$(-1) \times (-1) = 1 \notin G$

Hence, G does not possess a closure property w.r.t multiplication because all the multiplications do not belong to the set G

(iv) $\{1, -1\}$

*LHR 2022, GRW 2021-23, DGK 2022, RWP 2022, FSD 2021***Solution:**

Let $G = \{1, -1\}$

$1+1=2 \notin G$

$1+(-1)=(-1)+1=0 \notin G$

$-1+(-1)=-2 \notin G$

Hence, G does not possess a closure property w.r.t addition, because all the sums do not belong to the set G .

As $1 \times 1 = 1 \in G$

$1 \times (-1) = (-1) \times 1 = -1 \in G$

$(-1) \times (-1) = 1 \in G$

Hence, G possess a closure property w.r.t multiplications because all the products belong to the set G .

Q.2 Name the properties used in the following equations.

(Letters, where used, represent real numbers).

(i) $4+9=9+4$ *FSD2019*

Solution:

[Commutative Property w.r.t addition]

(ii) $(a+1)+\frac{3}{4}=a+\left(1+\frac{3}{4}\right)$

Solution:

[Associative Property w.r.t addition.]

(iii) $(\sqrt{3}+\sqrt{5})+\sqrt{7}=\sqrt{3}+(\sqrt{5}+\sqrt{7})$

Solution:

[Associative Property w.r.t addition.]

(iv) $100+0=100$

Solution:

[Additive Identity]

(v) $1000 \times 1 = 1000$ *FSD2019, RWP2023*

Solution:

[Multiplicative Identity]

(v) $4.1+(-4.1)=0$

Solution:

[Additive Inverse]

(vi) $a-a=0$

Solution:

[Additive Inverse]

(vii) $\sqrt{2} \times \sqrt{5} = \sqrt{5} \times \sqrt{2}$

Solution:

[Commutative property w.r.t multiplication.]

(viii) $a(b-c)=ab-ac$

Solution:

[Distributivity of multiplication over subtraction]

(ix) $(x-y)z=xz-yz$

Solution:

[Distributivity of multiplication over subtraction]

(x) $4 \times (5 \times 8) = (4 \times 5) \times 8$

Solution:

[Associative property w.r.t multiplication.]

(xi) $a(b+c-d)=ab+ac-ad$

Solution:

[Distributivity of multiplication over addition and subtraction.]

Q.3 Name the properties used in the following Inequalities:

(i) $-3 < -2 \Rightarrow 0 < 1$

Solution:

$-3 < -2$

By adding 3 on both sides

$-3+(3) < -2+3$

$0 < 1$

Additive property of inequalities

(ii) $-5 < -4 \Rightarrow 20 > 16$

Solution:

$-5 < -4$

By multiplying -4 on both sides

$(-4)(-5) > (-4)(-4)$

$20 > 16$

Multiplicative property of inequalities

(iii) $1 > -1 \Rightarrow -3 > -5$

Solution:

$1 > -1$

By adding (-4) on both sides

$1+(-4) > -1+(-4)$

$-3 > -5$

Additive property of inequalities

(iv) $a < 0 \Rightarrow -a > 0$

Solution:

$$a < 0$$

By multiplying -1 on both sides

$$(-1)a > (-1)0$$

$$-a > 0$$

Multiplicative property of inequalities

(v) $a > b \Rightarrow \frac{1}{a} < \frac{1}{b}$

Solution:

$$a > b$$

Multiplying both sides by $\frac{1}{ab}$

$$\frac{1}{ab}a > \frac{1}{ab}b$$

$$\frac{1}{b}\left(\frac{1}{a}a\right) > \frac{1}{a}\left(\frac{1}{b}b\right)$$

$$\frac{1}{b}(1) > \frac{1}{a}(1)$$

By multiplicative Inverse Law

$$\frac{1}{b} > \frac{1}{a}$$

$$\Rightarrow \frac{1}{a} < \frac{1}{b}$$

Multiplicative property of inequalities

(vi) $a > b \Rightarrow -a < -b$

Solution:

$$a > b$$

By multiplying both sides by -1

$$(-1)a < (-1)b$$

$$-a < -b$$

Q.4 Prove the following rules of addition.

(i) $\frac{a}{c} + \frac{b}{c} = \frac{a+b}{c}$

LHR 2019, GRW 2021, RWP 2021

Proof:

$$\text{L.H.S} = \frac{a}{c} + \frac{b}{c}$$

$$= a \cdot \frac{1}{c} + b \cdot \frac{1}{c} \quad [\text{Rule for product of fractions}]$$

$$= (a+b) \frac{1}{c} \quad [\text{Distributive Law}]$$

$$= \frac{a+b}{c} \quad [\text{Rule for product of fractions}]$$

$$= \text{R.H.S}$$

(ii) $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$

Proof:

$$\text{L.H.S} = \frac{a}{b} + \frac{c}{d}$$

$$= \frac{ad}{bd} + \frac{bc}{bd} \quad [\text{By Golden rule of fraction}]$$

$$= ad \cdot \frac{1}{bd} + bc \cdot \frac{1}{bd}$$

[Rule for product of fractions]

$$= (ad+bc) \cdot \frac{1}{bd} \quad [\text{Distributive Law}]$$

$$= \frac{ad+bc}{bd} \quad [\text{Rule for product of fractions}]$$

$$= \text{R.H.S}$$

Q.5 Prove that $-\frac{7}{12} - \frac{5}{18} = \frac{-21-10}{36}$

SGD 2021

Proof:

$$\text{L.H.S} = -\frac{7}{12} - \frac{5}{18}$$

$$= -\frac{7 \times 3}{12 \times 3} - \frac{5 \times 2}{18 \times 2}$$

[Golden rule of fraction]

$$= -\frac{21}{36} - \frac{10}{36}$$

$$= -21 \times \frac{1}{36} - 10 \times \frac{1}{36}$$

[Rule for product of fraction]

$$= (-21-10) \times \frac{1}{36} \quad [\text{Distributive Law}]$$

$$= \frac{-21-10}{36} \quad [\text{Rule for product of Fraction}]$$

$$= \text{R. H.S}$$

Q.6 Simplify by justifying each step:

(i) $\frac{4+16x}{4}$

Solution:

$$\frac{4+16x}{4}$$

$$= \frac{1}{4} \cdot (4+16x)$$

[Rule for product of Fraction]

$$= \frac{1}{4} \cdot [(4)(1) + (4)(4x)]$$

$$= \frac{1}{4} \cdot 4[1 + 4x] \quad [\text{Distributive Law}]$$

$$= 1[1 + 4x] \quad [\text{Multiplicative inverse}]$$

$$= 1 + 4x \quad [\text{Multiplicative Identity}]$$

$$(ii) \quad \frac{\frac{1}{4} + \frac{1}{5}}{\frac{1}{4} - \frac{1}{5}} \quad (\text{GRW 2018, RWP 2019, SHW, 2022})$$

Solution:

$$\frac{\frac{1}{4} + \frac{1}{5}}{\frac{1}{4} - \frac{1}{5}}$$

$$= \frac{\frac{1 \times 5}{4 \times 5} + \frac{1 \times 4}{5 \times 4}}{\frac{1 \times 5}{4 \times 5} - \frac{1 \times 4}{5 \times 4}} \quad [\text{Golden rule of fraction}]$$

$$= \frac{\frac{5}{20} + \frac{4}{20}}{\frac{5}{20} - \frac{4}{20}} \quad [\text{Closure Law}]$$

$$= \frac{5 \times \frac{1}{20} + 4 \times \frac{1}{20}}{5 \times \frac{1}{20} - 4 \times \frac{1}{20}} \quad [\text{Rule for product of fractions}]$$

$$= \frac{(5+4) \cdot \frac{1}{20}}{(5-4) \cdot \frac{1}{20}} \quad [\text{Distributive law}]$$

$$= \frac{(5+4) \cdot \frac{1}{20} \cdot 20}{(5-4) \cdot \frac{1}{20} \cdot 20} \quad [\text{Golden rule of Fraction}]$$

$$= \frac{(5+4) \cdot 1}{(5-4) \cdot 1} \quad [\text{Multiplicative Inverse}]$$

$$= \frac{9}{1} = 9 \quad [\text{Closure Law}]$$

$$(iii) \quad \frac{\frac{a}{b} + \frac{c}{d}}{\frac{a}{b} - \frac{c}{d}} \quad \text{FSD 2018, DGK 2022, SGD 2023}$$

Solution:

$$\frac{\frac{a}{b} + \frac{c}{d}}{\frac{a}{b} - \frac{c}{d}}$$

$$= \frac{\frac{ad}{bd} + \frac{bc}{bd}}{\frac{ad}{bd} - \frac{bc}{bd}} \quad [\text{Golden rule of fraction}]$$

$$= \frac{ad \cdot \frac{1}{bd} + bc \cdot \frac{1}{bd}}{ad \cdot \frac{1}{bd} - bc \cdot \frac{1}{bd}} \quad [\text{Rule for product of fractions}]$$

$$= \frac{(ad + bc) \cdot \frac{1}{bd}}{(ad - bc) \cdot \frac{1}{bd}} \quad [\text{Distributive Law}]$$

$$= \frac{(ad + bc) \cdot \frac{1}{bd} \cdot bd}{(ad - bc) \cdot \frac{1}{bd} \cdot bd} \quad [\text{Golden rule of fraction}]$$

$$= \frac{(ad + bc) \cdot 1}{(ad - bc) \cdot 1} \quad [\text{Multiplicative Inverse}]$$

$$= \frac{ad + bc}{ad - bc} \quad [\text{Multiplicative Identity}]$$

$$(iv) \quad \frac{\frac{1}{a} - \frac{1}{b}}{1 - \frac{1}{a} \cdot \frac{1}{b}} \quad (\text{RWP 2017})$$

Solution:

$$\frac{\frac{1}{a} - \frac{1}{b}}{1 - \frac{1}{a} \cdot \frac{1}{b}}$$

$$= \frac{\frac{1 \times b}{1 \times ab} - \frac{a \times 1}{a \times b}}{\frac{1 \times ab}{1 \times ab} - \frac{1}{ab}} \quad [\text{Golden rule of fraction}]$$

$$\begin{aligned}
 &= \frac{\frac{b}{ab} - \frac{a}{ab}}{\frac{ab}{ab} - \frac{1}{ab}} \quad [\text{Closure Law}] \\
 &= \frac{b \times \frac{1}{ab} - a \times \frac{1}{ab}}{ab \times \frac{1}{ab} - 1 \times \frac{1}{ab}} \\
 &\quad [\text{Rule for product of fractions}] \\
 &= \frac{(b-a) \cdot \frac{1}{ab}}{(ab-1) \cdot \frac{1}{ab}} \quad [\text{By Distributive law}]
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{(b-a) \cdot \frac{1}{ab} \cdot ab}{(ab-1) \cdot \frac{1}{ab} \cdot ab} \quad [\text{Golden rule of fraction}] \\
 &= \frac{(b-a) \cdot 1}{(ab-1) \cdot 1} \quad [\text{Multiplicative inverse}] \\
 &= \frac{b-a}{ab-1} \quad [\text{Multiplicative identity}]
 \end{aligned}$$

EXERCISE 1.2

Q.1 Verify the addition properties of complex numbers

(i) Closure Property: $\forall z_1, z_2 \in \mathbb{C}$

Solution:

$$\begin{aligned}
 \text{Let } z_1 &= a + bi, z_2 = c + di \\
 z_1 + z_2 &= (a + bi) + (c + di) \\
 &= (a + c) + (b + d)i \\
 &\Rightarrow z_1 + z_2 \in \mathbb{C}
 \end{aligned}$$

Hence, Closure property holds in complex numbers

(ii) Associative Property: $\forall z_1, z_2, z_3 \in \mathbb{C}$

Solution:

$$\begin{aligned}
 \text{Let } z_1 &= a + bi, z_2 = c + di, z_3 = e + fi \\
 z_1 + (z_2 + z_3) &= (z_1 + z_2) + z_3
 \end{aligned}$$

$$\begin{aligned}
 \text{L.H.S} &= z_1 + (z_2 + z_3) \\
 &= (a + bi) + [(c + di) + (e + fi)] \\
 &= (a + bi) + [(c + e) + (d + f)i] \\
 &= [a + (c + e)] + [b + (d + f)]i \\
 &= [(a + c) + e] + [(b + d) + f]i \\
 &= [(a + c) + (b + d)i] + (e + fi) \\
 &= [(a + bi) + (c + di)] + (e + fi) \\
 &= (z_1 + z_2) + z_3 \\
 &= \text{R. H. S}
 \end{aligned}$$

(iii) Identify Element: $\forall z \in \mathbb{C}, \exists 0 \in \mathbb{C}$

Solution:

$$\begin{aligned}
 \text{Let } z &= a + bi \\
 (a + bi) + (0 + i0) &= (0 + i0) + (a + bi) = a + bi
 \end{aligned}$$

(iv) Inverse element: $\forall z \in \mathbb{C}, \exists -z \in \mathbb{C}$

Solution:

$$\begin{aligned}
 \text{Let } z &= a + bi \Rightarrow -z = -a - bi \\
 (a + bi) + (-a - bi) &= (-a - bi) + (a + bi) = 0
 \end{aligned}$$

(v) Commutative Property: $\forall z_1, z_2 \in \mathbb{C}$

Solution:

$$\begin{aligned}
 \text{Let } z_1 &= a + bi, z_2 = c + di \\
 z_1 + z_2 &= (a + bi) + (c + di) \\
 &= (a + c) + (b + d)i \\
 &= (c + a) + (d + b)i \\
 &= (c + di) + (a + bi) \\
 &= z_2 + z_1
 \end{aligned}$$

Q.2 Verify the multiplication properties of Complex numbers.

(i) Closure Property: $\forall z_1, z_2 \in \mathbb{C}$

Solution:

$$\begin{aligned}
 \text{Let } z_1 &= a + bi, z_2 = c + di \\
 z_1 \cdot z_2 &= (a + bi)(c + di) \\
 &= ac + adi + bci + bdi^2 \\
 &= ac + (ad + bc)i - bd \\
 &= (ac - bd) + (ad + bc)i \\
 &\Rightarrow z_1 \cdot z_2 \in \mathbb{C}
 \end{aligned}$$

(ii) Associative Property: $\forall z_1, z_2, z_3 \in \mathbb{C}$

Solution:

$$\begin{aligned}
 z_1 &= a + bi, z_2 = c + di, z_3 = e + fi \\
 z_1 \cdot (z_2 \cdot z_3) &= (z_1 \cdot z_2) \cdot z_3 \\
 \text{L.H.S} &= z_1 \cdot (z_2 \cdot z_3) \\
 &= (a + bi) \cdot [(c + di) \cdot (e + fi)]
 \end{aligned}$$

$$\begin{aligned}
&= (a+bi) \cdot [ce + cfi + edi + dfi^2] \because i^2 = -1 \\
&= (a+bi) \cdot [ce + (cf + ed)i - df] \\
&= (a+bi) \cdot [(ce - df) + (cf + ed)i] \\
&= a(ce - df) + a(cf + ed)i + b(ce - df)i + b(cf + ed)i^2 \\
&= a(ce - df) + a(cf + ed)i + b(ce - df)i - b(cf + ed) \\
&= [a(ce - df) - b(cf + ed)] + [a(cf + ed) + b(ce - df)]i \\
&= (ace - adf - bcf - bde) + (acf + ade + bce - bdf)i \quad (i)
\end{aligned}$$

$$\text{R.H.S} = (z_1 \cdot z_2) \cdot z_3$$

$$\begin{aligned}
&= [(a+bi)(c+di)](e+fi) \\
&= [ac + adi + bci + bdi^2] \cdot (e+fi) \because i^2 = -1 \\
&= [ac + (ad + bc)i - bd] \cdot (e+fi) \\
&= [(ac - bd) + (ad + bc)i] \cdot (e+fi) \\
&= [(ac - bd)e - (ad + bc)f] + [(ac - bd)f + (ad + bc)e]i \\
&= (ace - bde - adf - bcf) + (acf - bdf + ade + bce)i \\
&= (ace - adf - bcf - bde) + (acf + ade + bce - bdf)i \quad (ii)
\end{aligned}$$

From equation (i) and equation (ii)

$$z_1 \cdot (z_2 \cdot z_3) = (z_1 \cdot z_2) \cdot z_3$$

(iii) **Identity Element:** $\forall z \in \mathbb{C}, \exists 1+0i \in \mathbb{C}$

Solution:

$$\text{Let } z = a + bi$$

$$(a+bi)(1+0i) = (1+0i)(a+bi) = a+bi$$

Hence, $1+0i$ is the multiplicative identity of complex number.

(iv) **Inverse Element:** $\forall z \in \mathbb{C}, \exists \frac{1}{z} \in \mathbb{C}$

with ($z \neq 0$)

Solution:

$$\text{Let } z = a + bi$$

$$\frac{1}{z} = \frac{1}{a+bi}$$

$$\begin{aligned}
&= \frac{1}{a+bi} \times \frac{a-bi}{a-bi} \\
&= \frac{a-bi}{a^2 - (bi)^2} \because i^2 = -1 \\
&= \frac{a-bi}{a^2 - b^2(-1)} \\
&= \frac{a-bi}{a^2 + b^2}
\end{aligned}$$

$$\frac{1}{z} = \left(\frac{a}{a^2 + b^2} \right) + \left(\frac{-b}{a^2 + b^2} \right) i$$

Now, we will be show that

$$z \cdot \frac{1}{z} = \frac{1}{z} \cdot z = 1$$

$$\begin{aligned}
z \cdot \frac{1}{z} &= (a+bi) \cdot \left(\frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2} i \right) \\
&= \frac{(a)^2 - (bi)^2}{a^2 + b^2} \because i^2 = -1 \\
&= \frac{a^2 - b^2(-1)}{a^2 + b^2} \\
&= \frac{a^2 + b^2}{a^2 + b^2} \\
&= 1
\end{aligned}$$

Similarly, $\frac{1}{z} \cdot z = 1$.

(v) **Commutative Property:** $\forall z_1, z_2 \in \mathbb{C}$

Solution:

$$\text{Let } z_1 = a + bi, z_2 = c + di$$

$$z_1 \cdot z_2 = z_2 \cdot z_1$$

$$\text{L.H.S} = z_1 \cdot z_2$$

$$\begin{aligned}
&= (a+bi)(c+di) \\
&= ac + adi + bci + bdi^2 \because i^2 = -1 \\
&= (ac - bd) + (ad + bc)i \quad (i)
\end{aligned}$$

$$\text{R.H.S} = z_2 \cdot z_1$$

$$\begin{aligned}
&= (c+di)(a+bi) \\
&= ca + cbi + dai + dbi^2 \because i^2 = -1 \\
&= ac + bci + adi + bd(-1) \\
&= (ac - bd) + (ad + bc)i \quad (ii)
\end{aligned}$$

By equation (i) and equation (ii) L.H.S = R. H. S

Q.3 Verify the distributive law of complex numbers

$$(a,b)[(c,d) + (e,f)] = (a,b) \cdot (c,d) + (a,b)(e,f)$$

Proof:

$$\begin{aligned}
\text{L.H.S} &= (a,b)[(c,d) + (e,f)] \\
&= (a,b) \cdot (c+e, d+f) \\
&\therefore (a,b)(c,d) = (ac - bd, ad + bc) \\
&= (a(c+e) - b(d+f), a(d+f) + b(c+e)) \\
&= (ac + ae + bd - bf, ad + af + bc + be) \quad (i)
\end{aligned}$$

$$\begin{aligned}
 \text{R.H.S} &= (a,b)(c,d) + (a,b)(e,f) \\
 &= (ac - bd, ad + bc) + (ae - bf, af + be) \\
 &= (ac - bd + ae - bf, ad + bc + af + be) \quad (\text{ii})
 \end{aligned}$$

By equation (i) and equation (ii) L.H.S = R. H. S

Q.4 Simplify the following.

(i) i^9 RWP 2021

Solution:

$$\begin{aligned}
 &i^9 \\
 &= i^8 \cdot i \\
 &= (i^2)^4 \cdot i \\
 &= (-1)^4 \cdot i \\
 &= 1 \cdot i \\
 &= i \quad \text{Answer}
 \end{aligned}$$

(ii) i^{14}

Solution:

$$\begin{aligned}
 &i^{14} \\
 &= (i^2)^7 \\
 &= (-1)^7 \\
 &= -1 \quad \text{Answer}
 \end{aligned}$$

(iii) $(-i)^{19}$

Solution:

$$\begin{aligned}
 &(-i)^{19} \\
 &= (-1 \times i)^{19} \\
 &= (-1)^{19} i^{19} \\
 &= (-1) i^{18} \cdot i \\
 &= (-1) (i^2)^9 \cdot i \\
 &= (-1) (-1)^9 \cdot i \\
 &= (-1) (-1) i \\
 &= i \quad \text{Answer}
 \end{aligned}$$

(iv) $(-1)^{\frac{-21}{2}}$

LHR 2018, FSD 21

Solution:

$$\begin{aligned}
 &(-1)^{\frac{-21}{2}} \\
 &= [i^2]^{-\frac{21}{2}} \\
 &= i^{-21} \\
 &= \frac{1}{i^{21}}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{i^{20} \cdot i} \\
 &= \frac{1}{(i^2)^{10} \cdot i} \\
 &= \frac{1}{(-1)^{10} \cdot i}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{i} \\
 &= -i \quad \text{Answer}
 \end{aligned}$$

Q.5 Write in terms of i .

(i) $\sqrt{-1} b$

Solution:

$$\begin{aligned}
 &= \sqrt{-1} b \\
 &= ib
 \end{aligned}$$

(ii) $\sqrt{-5}$

Solution:

$$\begin{aligned}
 &= \sqrt{-5} \\
 &= \sqrt{-1 \times 5} \\
 &= \sqrt{-1} \sqrt{5} \\
 &= i\sqrt{5} \quad \text{Answer}
 \end{aligned}$$

(iii) $\sqrt{\frac{-16}{25}}$

Solution:

$$\begin{aligned}
 &= \sqrt{\frac{-16}{25}} \\
 &= \sqrt{\frac{-1 \times 16}{25}} \\
 &= \frac{\sqrt{-1} \sqrt{16}}{\sqrt{25}} \\
 &= \frac{4}{5} i \quad \text{Answer}
 \end{aligned}$$

(iv) $\sqrt{\frac{1}{-4}}$

Solution:

$$\begin{aligned}
&= \sqrt{\frac{1}{-4}} \\
&= \frac{1}{\sqrt{-1 \times 4}} \\
&= \frac{1}{\sqrt{-1} \sqrt{4}} \\
&= \frac{1}{2i} \\
&= \frac{1}{2i} \times \frac{i}{i} \\
&= \frac{i}{2i^2} \\
&= \frac{i}{2(-1)} \\
&= \frac{-i}{2} \text{ Answer}
\end{aligned}$$

Simplify the following:

Q.6 $(7,9) + (3,-5)$

Solution:

$$\begin{aligned}
&= (7,9) + (3,-5) \\
&= (7+3, 9+(-5)) \\
&= (10,4) \text{ Answer}
\end{aligned}$$

Q.7 $(8,-5) - (-7,4)$

Solution:

$$\begin{aligned}
&= (8,-5) - (-7,4) \\
&= (8-(-7), -5-4) \\
&= (15,-9) \text{ Answer}
\end{aligned}$$

Q.8 $(2,6)(3,7)$

Solution:

$$\begin{aligned}
&= (2,6)(3,7) \\
&= (2+6i)(3+7i) \\
&= 6+14i+18i+42i^2 \\
&= 6+32i+42(-1) \\
&= 6-42+32i \\
&= -36+32i \\
&= (-36,32) \text{ Answer}
\end{aligned}$$

Q.9 $(5,-4)(-3,-2)$

LHR 2022, MTN 2023, GRW 2023, SGD 2021

Solution:

$$\begin{aligned}
&= (5,-4)(-3,-2) \\
&= (5-4i)(-3-2i) \\
&= -15-10i+12i+8i^2 \\
&= -15+2i+8(-1) \\
&= -15+2i-8 \\
&= -23+2i \\
&= (-23,2) \text{ Answer}
\end{aligned}$$

Q.10 $(0,3)(0,5)$

Solution:

$$\begin{aligned}
&= (0,3)(0,5) \\
&= (0+3i)(0+5i) \\
&= (3i)(5i) \\
&= 15i^2 \\
&= 15(-1) \\
&= -15+0i \\
&= (-15,0) \text{ Answer}
\end{aligned}$$

Q.11 $(2,6) \div (3,7)$ *GRW 2022, MTN 2022*

Solution:

$$\begin{aligned}
&= (2,6) \div (3,7) \\
&= \frac{2+6i}{3+7i} \\
&= \frac{2+6i}{3+7i} \times \frac{3-7i}{3-7i} \\
&= \frac{(2+6i)(3-7i)}{(3+7i)(3-7i)} \\
&= \frac{6-14i+18i-42i^2}{3^2 - (7i)^2} \\
&= \frac{6+4i-42(-1)}{9-49(-1)} \\
&= \frac{6+4i+42}{9+49} \\
&= \frac{48+4i}{58} \\
&= \frac{48}{58} + \frac{4}{58}i
\end{aligned}$$

$$= \frac{24}{29} + \frac{2}{29}i$$

$$= \left(\frac{24}{29}, \frac{2}{29} \right) \text{ Answer}$$

Q.12 $(5, -4) \div (-3, -8)$ *LHR 2021*

Solution:

$$= (5, -4) \div (-3, -8)$$

$$= \frac{5 - 4i}{-3 - 8i}$$

$$= \frac{5 - 4i}{-3 - 8i} \times \frac{-3 + 8i}{-3 + 8i}$$

$$= \frac{-15 + 40i + 12i - 32i^2}{(-3)^2 - (8i)^2}$$

$$= \frac{-15 + 52i - 32(-1)}{9 - 64(-1)}$$

$$= \frac{-15 + 52i + 32}{9 + 64}$$

$$= \frac{17 + 52i}{73}$$

$$= \frac{17}{73} + \frac{52}{73}i$$

$$= \left(\frac{17}{73}, \frac{52}{73} \right) \text{ Answer}$$

Q.13 Prove that the sum as well as the product of any two conjugate complex numbers is a real number. *FSD 2022, GRW 2019*

Proof:

Let $z = x + iy$ and $\bar{z} = x - iy$

$$z + \bar{z} = x + i\cancel{y} + x - i\cancel{y}$$

$$= 2x \text{ is a real number}$$

$$z \cdot \bar{z} = (x + iy)(x - iy)$$

$$= (x)^2 - (iy)^2 \because a^2 - b^2 = (a + b)(a - b)$$

$$= x^2 - i^2 y^2$$

$$= x^2 - (-1) y^2$$

$$= x^2 + y^2 \text{ is a real number.}$$

Q.14 Find the multiplicative inverse of each of the following numbers:

(i) $(-4, 7)$

LHR 2018, SGD 2021-22, RWP 2023

Solution:

Let $z = (-4, 7)$

$$z = -4 + 7i$$

$$\frac{1}{z} = \frac{1}{-4 + 7i}$$

$$= \frac{1}{-4 + 7i} \times \frac{-4 - 7i}{-4 - 7i}$$

$$= \frac{-4 - 7i}{(-4)^2 - (7i)^2}$$

$$= \frac{-4 - 7i}{16 - 49(-1)}$$

$$= \frac{-4 - 7i}{16 + 49}$$

$$= \frac{-4 - 7i}{65}$$

$$= \frac{-4}{65} - \frac{7}{65}i$$

$$\frac{1}{z} = \left(\frac{-4}{65}, \frac{-7}{65} \right) \text{ Answer}$$

(ii) $(\sqrt{2}, -\sqrt{5})$

GRW 2021-22, LHR 2019-22, FSD 2021

Solution:

Let $z = (\sqrt{2}, -\sqrt{5})$

$$z = \sqrt{2} - \sqrt{5}i$$

$$\frac{1}{z} = \frac{1}{\sqrt{2} - \sqrt{5}i}$$

$$= \frac{1}{\sqrt{2} - \sqrt{5}i} \times \frac{\sqrt{2} + \sqrt{5}i}{\sqrt{2} + \sqrt{5}i}$$

$$= \frac{\sqrt{2} + \sqrt{5}i}{(\sqrt{2})^2 - (\sqrt{5}i)^2}$$

$$\frac{1}{z} = \frac{\sqrt{2} + \sqrt{5}i}{2 - 5(-1)}$$

$$= \frac{\sqrt{2} + \sqrt{5}i}{2 + 5}$$

$$= \frac{\sqrt{2} + \sqrt{5}i}{7}$$

$$= \frac{\sqrt{2}}{7} + \frac{\sqrt{5}}{7}i$$

$$\frac{1}{z} = \left(\frac{\sqrt{2}}{7}, \frac{\sqrt{5}}{7} \right) \text{ Answer}$$

(iii) (1, 0)

Solution:

Let $z = (1, 0)$

$z = 1 + 0i$

$$\frac{1}{z} = \frac{1}{1+0i}$$

$$\frac{1}{z} = \frac{1}{1+0i} \times \frac{1-0i}{1-0i}$$

$$= \frac{1-0i}{(1)^2 - (0i)^2}$$

$$= \frac{1}{1}$$

$$\frac{1}{z} = 1$$

$$\frac{1}{z} = (1, 0) \text{ Answer}$$

Q.15 Factorize the following(i) $a^2 + 4b^2$ *LHR 2019, GRW 2021, SGD 2018***Solution:**

$$= a^2 + 4b^2$$

$$= a^2 - (-4b^2)$$

$$= a^2 - (-1)4b^2$$

$$= a^2 - i^2 4b^2$$

$$= a^2 - (i2b)^2$$

$$= (a - i2b)(a + i2b) \text{ Answer}$$

(ii) $9a^2 + 16b^2$ *FSD 2018-22, LHR 2022, DGK 2022, GRW 2023***Solution:**

$$9a^2 + 16b^2$$

$$= 9a^2 - (-16b^2)$$

$$= 9a^2 - (-1)16b^2$$

$$= 9a^2 - i^2 16b^2$$

$$= (3a)^2 - (i4b)^2$$

$$= (3a - i4b)(3a + i4b) \text{ Answer}$$

(iii) $3x^2 + 3y^2$ *SHW 2023***Solution:**

$$= 3x^2 + 3y^2$$

$$= 3[x^2 + y^2]$$

$$= 3[x^2 - (-y^2)]$$

$$= 3[x^2 - (-1)y^2]$$

$$= 3[x^2 - i^2 y^2]$$

$$= 3[x^2 - (iy)^2]$$

$$= 3(x - iy)(x + iy) \text{ Answer}$$

Q.16 Separate into real and imaginary parts (Write as a simple complex number):(i) $\frac{2-7i}{4+5i}$ *LHR 2021, SGD 2022***Solution:**

$$= \frac{2-7i}{4+5i}$$

$$= \frac{2-7i}{4+5i} \times \frac{4-5i}{4-5i}$$

$$= \frac{(2-7i)(4-5i)}{(4)^2 - (5i)^2}$$

$$= \frac{8-10i-28i+35i^2}{16-25(-1)}$$

$$= \frac{8-38i+35(-1)}{16+25}$$

$$= \frac{8-38i-35}{41}$$

$$= \frac{-27-38i}{41}$$

$$= \frac{-27}{41} - \frac{38}{41}i$$

$$\text{Real Part} = \frac{-27}{41}, \text{ Imaginary Part} = \frac{-38}{41}$$

(ii) $\frac{(-2+3i)^2}{1+i}$ *DGK 2022***Solution:**

$$= \frac{(-2+3i)^2}{1+i}$$

$$\begin{aligned}
 &= \frac{(-2)^2 + (3i)^2 + 2(-2)(3i)}{1+i} \\
 &= \frac{4-9-12i}{1+i} \\
 &= \frac{-5-12i}{1+i} \\
 &= \frac{-5-12i}{1+i} \times \frac{1-i}{1-i} \\
 &= \frac{(-5-12i)(1-i)}{(1+i)(1-i)} \\
 &= \frac{-5+5i-12i+12i^2}{1^2-i^2} \\
 &= \frac{-5-7i+12(-1)}{1-(-1)} \\
 &= \frac{-5-7i-12}{1+1} \\
 &= \frac{-17-7i}{2} \\
 &= \frac{-17}{2} - \frac{7}{2}i \\
 \text{Real Part} &= \frac{-17}{2}, \text{ Imaginary Part} = \frac{-7}{2}
 \end{aligned}$$

(iii) $\frac{i}{1+i}$
GRW 2019, RWP 2018-23, MTN 2022-23, SHW 2022)

Solution:

$$\begin{aligned}
 &= \frac{i}{1+i} \\
 &= \frac{i}{1+i} \times \frac{1-i}{1-i} \\
 &= \frac{i(1-i)}{1^2-i^2} \\
 &= \frac{i-i^2}{1-(-1)} \\
 &= \frac{i-(-1)}{2} \\
 &= \frac{1+i}{2} \\
 &= \frac{1}{2} + \frac{1}{2}i \\
 \text{Real Part} &= \frac{1}{2}, \text{ Imaginary Part} = \frac{1}{2}
 \end{aligned}$$

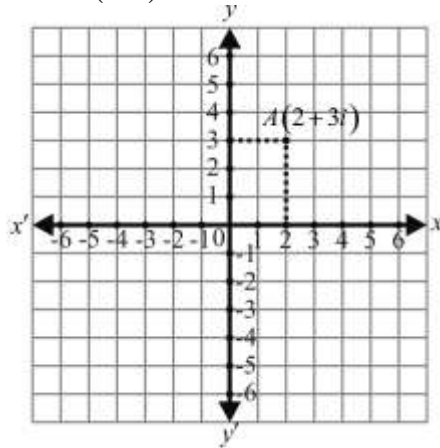
EXERCISE 1.3

Q.1 Graph the following numbers in the complex plane.

(i) $2+3i$

Solution:

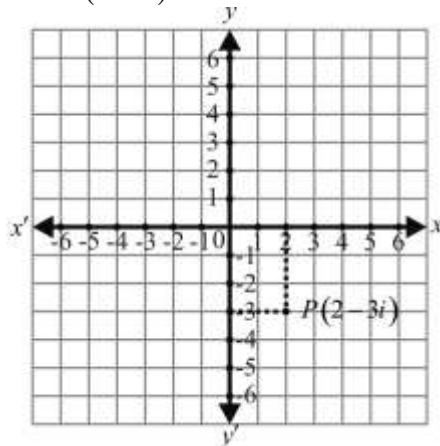
As $2+3i=(2,3)$



(ii) $2-3i$

Solution:

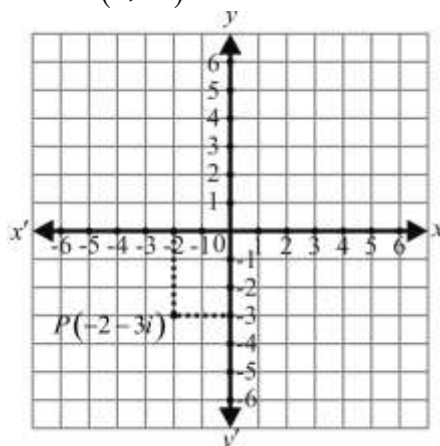
As $2-3i=(2,-3)$



(iii) $-2-3i$

Solution:

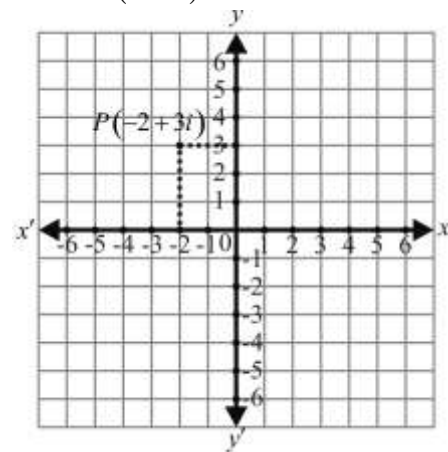
As $-2-3i=(-2,-3)$



(iv) $-2+3i$

Solution:

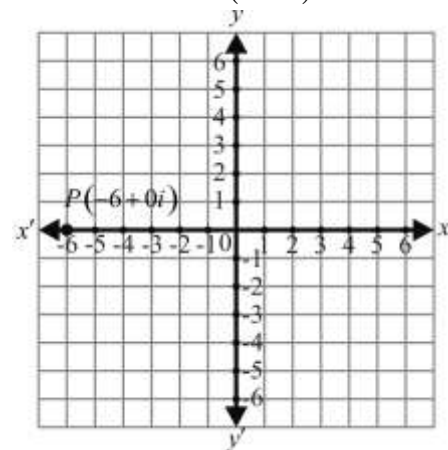
As $-2+3i=(-2,3)$



(v) -6

Solution:

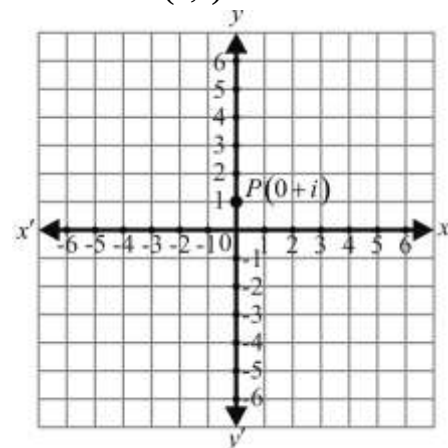
$-6=-6+0i=(-6,0)$



(vi) i

Solution:

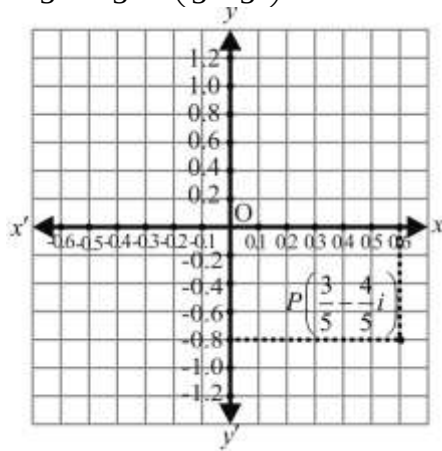
$i=0+i=(0,1)$



(vii) $\frac{3}{5} - \frac{4}{5}i$

Solution:

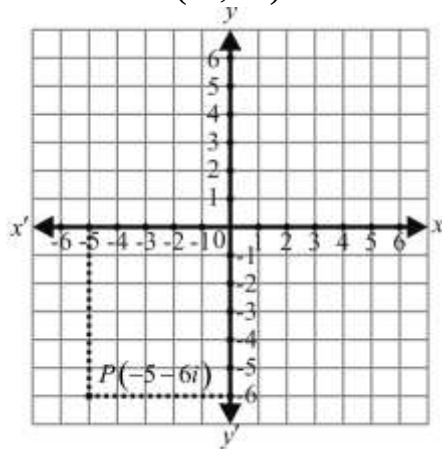
$$\frac{3}{5} - \frac{4}{5}i = \left(\frac{3}{5}, -\frac{4}{5}\right)$$



(viii) $-5 - 6i$

Solution:

$$-5 - 6i = (-5, -6)$$



Q.2 Find the multiplicative inverse of each of the following numbers:

(i) $-3i$

Solution:

Let $z = -3i$

$$\begin{aligned} \frac{1}{z} &= \frac{1}{-3i} \\ &= \frac{1}{-3i} \times \frac{3i}{3i} \\ &= \frac{3i}{-9i^2} \\ &= \frac{3i}{-9(-1)} \\ &= \frac{3i}{9} \end{aligned}$$

$$\frac{1}{z} = \frac{1}{3}i \text{ Answer}$$

(ii) $1 - 2i$

Solution:

Let $z = 1 - 2i$

$$\begin{aligned} \frac{1}{z} &= \frac{1}{1 - 2i} \\ &= \frac{1}{1 - 2i} \times \frac{1 + 2i}{1 + 2i} \\ &= \frac{1 + 2i}{(1)^2 - (2i)^2} \\ \because a^2 - b^2 &= (a + b)(a - b) \\ &= \frac{1 + 2i}{1 - 4(-1)} \end{aligned}$$

$$\frac{1}{z} = \frac{1 + 2i}{1 + 4}$$

$$\frac{1}{z} = \frac{1}{5} + \frac{2}{5}i \text{ Answer}$$

(iii) $-3 - 5i$

Solution:

Let $z = -3 - 5i$

$$\begin{aligned} \frac{1}{z} &= \frac{1}{-3 - 5i} \\ \frac{1}{z} &= \frac{1}{-3 - 5i} \times \frac{-3 + 5i}{-3 + 5i} \\ &= \frac{-3 + 5i}{(-3)^2 - (5i)^2} \\ \because a^2 - b^2 &= (a + b)(a - b) \\ &= \frac{-3 + 5i}{9 - 25i^2} \\ &= \frac{-3 + 5i}{9 - 25(-1)} \\ &= \frac{-3 + 5i}{9 + 25} \\ &= \frac{-3 + 5i}{34} \end{aligned}$$

$$\frac{1}{z} = \frac{-3}{34} + \frac{5}{34}i \text{ Answer}$$

(iv) (1, 2)

Solution:

Let $z = (1, 2)$

$$z = 1 + 2i$$

$$\frac{1}{z} = \frac{1}{1+2i}$$

$$= \frac{1}{1+2i} \times \frac{1-2i}{1-2i}$$

$$= \frac{1-2i}{(1)^2 - (2i)^2}$$

$$\because a^2 - b^2 = (a+b)(a-b)$$

$$= \frac{1-2i}{1-4(-1)}$$

$$= \frac{1-2i}{1+4}$$

$$= \frac{1-2i}{5}$$

$$= \frac{1}{5} - \frac{2}{5}i$$

$$\frac{1}{z} = \left(\frac{1}{5}, -\frac{2}{5} \right) \text{ Answer}$$

Q.3 Simplify

(i) i^{101}

GRW 2019

Solution:

$$= i^{101}$$

$$= i^{100} \cdot i$$

$$= (i^2)^{50} \cdot i$$

$$= (-1)^{50} \cdot i$$

$$= 1 \cdot i$$

$$= i \text{ Answer}$$

(ii) $(-ai)^4, a \in \mathbb{R}$

Solution:

$$= (-ai)^4$$

$$= (-a \times i)^4$$

$$= (-a)^4 i^4$$

$$= a^4 (i^2)^2$$

$$= a^4 (-1)^2$$

$$= a^4 \times 1$$

$$= a^4 \text{ Answer}$$

(iii) i^{-3}

Solution:

$$= i^{-3}$$

$$= \frac{1}{i^3}$$

$$= \frac{1}{i^2 \cdot i}$$

$$= \frac{1}{-i}$$

$$= \frac{1}{-i} \times \frac{i}{i}$$

$$= \frac{i}{-i^2}$$

$$= \frac{i}{-(-1)}$$

$$= i \text{ Answer}$$

(iv) i^{-10}

Solution:

$$= i^{-10}$$

$$= \frac{1}{i^{10}}$$

$$= \frac{1}{(i^2)^5}$$

$$= \frac{1}{(-1)^5}$$

$$= \frac{1}{-1}$$

$$= -1 \text{ Answer}$$

Q.4 Prove that $z = \bar{z}$ if z is real .

FSD 2019, GRW 2022, MTN 2022, RWP 2022-23, SHW 2023

Proof:

Let $z = x + iy$ (i)

Suppose that $z = \bar{z}$

Now we have to prove z is real

As $z = \bar{z}$

$$x + iy = x - iy$$

$$x + iy - x + iy = 0$$

$$2iy = 0$$

As $2i \neq 0, y = 0$

Put in a equation (i)

$$z = x + i(0)$$

$$z = x$$

Hence z is a real number.

Conversely,

Suppose z is a real number

Now we have to prove that $z = \bar{z}$

Let $z = x + 0i = x$

$$\bar{z} = x - 0i = x$$

Hence

$$z = \bar{z}$$

Q.5 Simplify by expressing in the form

$$a + bi$$

(i) $5 + 2\sqrt{-4}$

Solution:

$$= 5 + 2\sqrt{-4}$$

$$= 5 + 2\sqrt{-1 \times 4}$$

$$= 5 + 2\sqrt{-1} \sqrt{4}$$

$$= 5 + 2(i)2$$

$$= 5 + 4i \text{ Answer}$$

(ii) $(2 + \sqrt{-3})(3 + \sqrt{-3})$ GRW 2021

Solution:

$$= (2 + \sqrt{-3})(3 + \sqrt{-3})$$

$$= (2 + \sqrt{-1 \times 3})(3 + \sqrt{-1 \times 3})$$

$$= (2 + \sqrt{-1} \sqrt{3})(3 + \sqrt{-1} \sqrt{3})$$

$$= (2 + i\sqrt{3})(3 + i\sqrt{3})$$

$$= 6 + i2\sqrt{3} + i3\sqrt{3} + i^2(\sqrt{3})^2$$

$$= 6 + i5\sqrt{3} + (-1)(3)$$

$$= 6 + i5\sqrt{3} - 3$$

$$= 3 + i5\sqrt{3} \text{ Answer}$$

(iii) $\frac{2}{\sqrt{5} + \sqrt{-8}}$

Solution:

$$= \frac{2}{\sqrt{5} + \sqrt{-8}}$$

$$= \frac{2}{\sqrt{5} + \sqrt{-1 \times 8}}$$

$$= \frac{2}{\sqrt{5} + \sqrt{-1} \sqrt{8}}$$

$$= \frac{2}{\sqrt{5} + i\sqrt{8}}$$

$$= \frac{2}{\sqrt{5} + i\sqrt{8}} \times \frac{\sqrt{5} - i\sqrt{8}}{\sqrt{5} - i\sqrt{8}}$$

$$= \frac{2[\sqrt{5} - i\sqrt{8}]}{(\sqrt{5})^2 - (i\sqrt{8})^2}$$

$$\because a^2 - b^2 = (a+b)(a-b)$$

$$= \frac{2[\sqrt{5} - i\sqrt{8}]}{5 - (-1)8}$$

$$= \frac{2[\sqrt{5} - i\sqrt{8}]}{13}$$

$$= \frac{2\sqrt{5}}{13} - \frac{2\sqrt{8}}{13}i$$

$$= \frac{2\sqrt{5}}{13} - \frac{2 \times 2\sqrt{2}}{13}i$$

$$= \frac{2\sqrt{5}}{13} - \frac{4\sqrt{2}}{13}i \text{ Answer}$$

(iv) $\frac{3}{\sqrt{6} - \sqrt{-12}}$

Solution:

$$= \frac{3}{\sqrt{6} - \sqrt{-12}}$$

$$= \frac{3}{\sqrt{6} - \sqrt{-1 \times 12}}$$

$$= \frac{3}{\sqrt{6} - i\sqrt{12}}$$

$$= \frac{3}{\sqrt{6} - i\sqrt{12}} \times \frac{\sqrt{6} + i\sqrt{12}}{\sqrt{6} + i\sqrt{12}}$$

$$= \frac{3[\sqrt{6} + i\sqrt{12}]}{(\sqrt{6})^2 - (i\sqrt{12})^2}$$

$$\because a^2 - b^2 = (a+b)(a-b)$$

$$= \frac{3[\sqrt{6} + i\sqrt{12}]}{6 - (-1)12}$$

$$\begin{aligned}
 &= \frac{3[\sqrt{6} + i\sqrt{12}]}{18} \\
 &= \frac{\sqrt{6} + i\sqrt{12}}{6} \\
 &= \frac{\sqrt{6}}{6} + i \frac{\sqrt{6}\sqrt{2}}{6} \\
 &= \frac{1}{\sqrt{6}} + i \frac{\sqrt{2}}{\sqrt{6}} \\
 &= \frac{1}{\sqrt{6}} + i \frac{\sqrt{2}}{\sqrt{3}\sqrt{2}} \\
 &= \frac{1}{\sqrt{6}} + i \frac{1}{\sqrt{3}} \text{ Answer}
 \end{aligned}$$

Q.6 Show that $\forall z \in \mathbb{C}$

- (i) $z^2 + \bar{z}^2$ is a real number.
GRW 2019-21, FSD 2022, BWP 2023, LHR 2023

Solution:

Let $z = x + iy$

$$\bar{z} = x - iy$$

$$\begin{aligned}
 z^2 + \bar{z}^2 &= (x + iy)^2 + (x - iy)^2 \\
 &= x^2 + (iy)^2 + \cancel{2x(iy)} + x^2 + (iy)^2 - \cancel{2x(iy)} \\
 &= x^2 + (-1)y^2 + x^2 + (-1)y^2 \\
 &= 2x^2 - 2y^2 \text{ is a real number}
 \end{aligned}$$

- (ii) $(z - \bar{z})^2$
GRW 2018, RWP 2017, LHR 2022, MTN 2022

Solution:

Let $z = x + iy$

$$z = x - iy$$

$$\begin{aligned}
 (z - \bar{z}) &= (x + iy) - (x - iy) \\
 &= x + iy - x + iy
 \end{aligned}$$

$$z - \bar{z} = 2iy$$

By taking square on both sides

$$\begin{aligned}
 (z - \bar{z})^2 &= (2iy)^2 \\
 &= 4(-1)y^2 \\
 &= -4y^2 \text{ is a real number}
 \end{aligned}$$

Q.7 Simplify the following:

(i) $\left(\frac{-1}{2} + \frac{\sqrt{3}}{2}i\right)^3$

Solution:

$$\begin{aligned}
 &= \left(\frac{-1}{2} + \frac{\sqrt{3}}{2}i\right)^3 \\
 &= \left(\frac{-1 + \sqrt{3}i}{2}\right)^3 \\
 &= \frac{(-1 + \sqrt{3}i)^3}{2^3} \\
 &= \frac{1}{8}[-1 + \sqrt{3}i]^3 \\
 &= \frac{1}{8}[(-1)^3 + 3(-1)^2(\sqrt{3}i) + 3(-1)(\sqrt{3}i)^2 + (\sqrt{3}i)^3] \\
 &= \frac{1}{8}[-1 + 3\sqrt{3}i - 3 \times 3(-1) + (\sqrt{3})^3 i^3] \\
 &= \frac{1}{8}[-1 + 3\sqrt{3}i + 9 + (\sqrt{3})^2 \sqrt{3}(-i)] \\
 &= \frac{1}{8}[-1 + 3\sqrt{3}i + 9 - 3\sqrt{3}i] \\
 &= \frac{1}{8}[8] \\
 &= 1 \text{ Answer}
 \end{aligned}$$

(ii) $\left(\frac{-1}{2} - \frac{\sqrt{3}}{2}i\right)^3$ *RWP 2018, FSD 2022*

Solution:

$$\begin{aligned}
 &= \left(\frac{-1}{2} - \frac{\sqrt{3}}{2}i\right)^3 \\
 &= \left(\frac{-1 - \sqrt{3}i}{2}\right)^3 \\
 &= \frac{(-1 - \sqrt{3}i)^3}{2^3} \\
 &= \frac{1}{8}[-1 - \sqrt{3}i]^3 \\
 &= \frac{1}{8}[(-1)^3 - 3(-1)^2(\sqrt{3}i) + 3(-1)(\sqrt{3}i)^2 - (\sqrt{3}i)^3] \\
 &= \frac{1}{8}[-1 - 3\sqrt{3}i - 3 \times 3(-1) - (\sqrt{3})^3 i^3] \\
 &= \frac{1}{8}[-1 - 3\sqrt{3}i + 9 - (\sqrt{3})^2 \sqrt{3}(-i)]
 \end{aligned}$$

$$= \frac{1}{8} [-1 - 3\sqrt{3}i + 9 + 3\sqrt{3}i]$$

$$= \frac{1}{8} [8]$$

$$= 1 \text{ Answer}$$

$$(iii) \left(\frac{-1}{2} - \frac{\sqrt{3}}{2}i \right)^{-2} \left(\frac{-1}{2} - \frac{\sqrt{3}}{2}i \right)$$

Solution:

$$= \left(\frac{-1}{2} - \frac{\sqrt{3}}{2}i \right)^{-2} \left(\frac{-1}{2} - \frac{\sqrt{3}}{2}i \right)$$

$$= \left(\frac{-1}{2} - \frac{\sqrt{3}}{2}i \right)^{-2+1}$$

$$= \left(\frac{-1}{2} - \frac{\sqrt{3}}{2}i \right)^{-1}$$

$$= \left(\frac{-1 - \sqrt{3}i}{2} \right)^{-1}$$

$$= \frac{2}{-1 - \sqrt{3}i}$$

$$= \frac{2}{-1 - \sqrt{3}i} \times \frac{-1 + \sqrt{3}i}{-1 + \sqrt{3}i}$$

$$= \frac{2 [-1 + \sqrt{3}i]}{(-1)^2 - (\sqrt{3}i)^2}$$

$$= \frac{2 [-1 + \sqrt{3}i]}{1 - 3(-1)}$$

$$= \frac{2 [-1 + \sqrt{3}i]}{4}$$

$$= \frac{-1 + \sqrt{3}i}{2} \text{ Answer}$$

$$(iv) (a + bi)^2$$

Solution:

$$= (a + bi)^2$$

$$= a^2 + (bi)^2 + 2(a)(bi)$$

$$= a^2 + b^2(-1) + 2abi$$

$$= a^2 - b^2 + 2abi \text{ Answer}$$

$$(v) (a + bi)^{-2}$$

Solution:

$$= (a + bi)^{-2}$$

$$= \frac{1}{(a + bi)^2}$$

$$= \frac{1}{(a + bi)^2} \times \frac{(a - bi)^2}{(a - bi)^2}$$

$$= \frac{(a - bi)^2}{(a + bi)^2 (a - bi)^2}$$

$$= \frac{a^2 + (bi)^2 - 2(a)(bi)}{[a^2 - (bi)^2]^2}$$

$$= \frac{a^2 + b^2(-1) - 2abi}{[a^2 - b^2(-1)]^2}$$

$$= \frac{a^2 - b^2 - 2abi}{(a^2 + b^2)^2}$$

$$= \frac{a^2 - b^2}{(a^2 + b^2)^2} - \frac{2ab}{(a^2 + b^2)^2}i \text{ Answer}$$

$$(vi) (a + bi)^3 \quad \text{BWP 2022}$$

Solution:

$$= (a + bi)^3$$

$$= a^3 + 3(a)^2(bi) + 3a(bi)^2 + (bi)^3$$

$$= a^3 + 3a^2bi + 3ab^2(-1) + b^3i^3$$

$$= a^3 + 3a^2bi - 3ab^2 + b^3i^2.i$$

$$= a^3 + 3a^2bi - 3ab^2 + b^3(-1).i$$

$$= a^3 - 3ab^2 + 3a^2bi - b^3i$$

$$= (a^3 - 3ab^2) + (3a^2b - b^3)i \text{ Answer}$$

$$(vii) (a - bi)^3$$

Solution:

$$= (a - bi)^3$$

$$= a^3 - 3(a)^2(bi) + 3a(bi)^2 - (bi)^3$$

$$= a^3 - 3a^2bi + 3ab^2(-1) - b^3i^3$$

$$= a^3 - 3a^2bi - 3ab^2 - b^3i^2.i$$

$$= a^3 - 3a^2bi - 3ab^2 - b^3(-1).i$$

$$= a^3 - 3a^2bi - 3ab^2 + b^3i$$

$$= (a^3 - 3ab^2) + (-3a^2b + b^3)i \text{ Answer}$$

$$(viii) (3 - \sqrt{-4})^{-3}$$

Solution:

$$= (3 - \sqrt{-4})^{-3}$$

$$\begin{aligned}&= (3 - \sqrt{-1}\sqrt{4})^{-3} \\&= (3 - i \cdot 2)^{-3} \\&= \frac{1}{(3 - 2i)^3} \\&= \frac{1}{(3 - 2i)^3} \times \frac{(3 + 2i)^3}{(3 + 2i)^3} \\&= \frac{(3 + 2i)^3}{[(3 - 2i)(3 + 2i)]^3} \\&= \frac{3^3 + 3(3)^2(2i) + 3(3)(2i)^2 + (2i)^3}{[3^2 - (2i)^2]^3} \\&= \frac{27 + 54i + 9 \times 4(-1) + 8i^3}{[9 - 4(-1)]^3} \\&= \frac{27 + 54i - 36 + 8i^2i}{(13)^3} \\&= \frac{27 + 54i - 36 + 8(-1)i}{13^3} \\&= \frac{-9 + 54i - 8i}{2197} \\&= \frac{-9 + 46i}{2197} \\&= \frac{-9}{2197} + \frac{46}{2197}i \text{ Answer}\end{aligned}$$

SELF IMPROVEMENT TEST (SIT)**Q.No.1 Multiple Choice Questions.**

- 3.141592 is a/an_____ number:
 (A) Natural (B) Whole (C) Rational (D) Irrational
- If $x = y \wedge y = z \Rightarrow x = z, \forall x, y, z \in \mathbf{R}$ this property is called:
 (A) Symmetric (B) Transitive (C) Additive (D) Trichotomy
- The imaginary part of $z = 2 + \sqrt{-4}i$ is: **(CONCEPTUAL)**
 (A) 0 (B) 2 (C) $\sqrt{-4}$ (D) 4
- The multiplicative inverse of $z = -3 - i^{11}$ is: **(CONCEPTUAL)**
 (A) $\left(\frac{3}{10}, \frac{1}{10}\right)$ (B) $\left(-\frac{3}{10}, \frac{1}{10}\right)$ (C) $\left(-\frac{3}{10}, -\frac{1}{10}\right)$ (D) $\left(\frac{3}{10}, -\frac{1}{10}\right)$
- The modulus of $\frac{1+i}{1-i}$ is equal to: **(CONCEPTUAL)**
 (A) 1 (B) 2 (C) -1 (D) 0
- If $z = 1 + i\sqrt{3}$ then argument of z is:
 (A) 30° (B) 45° (C) 60° (D) 90°
- The complex number $z = \frac{3-4i}{5}$ lies in_____ quadrant:
 (A) 1st (B) 2nd (C) 3rd (D) 4th

Q.No.2 Short Questions.

- Prove by rules of addition $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$
- Express $\frac{2-7i}{4+5i}$ in the form of $a+bi$.
- Factorize: $3x^2 + 5y^2$
- If $z = \cos \theta + i \sin \theta$ then find the value of $z - \frac{1}{z}$ **(CONCEPTUAL)**

Q.No.3 Long Questions.

- Prove that $\sqrt{3}$ is an irrational number.
- Simplify: $\left(\frac{i+\sqrt{3}}{2}\right)^3 + \left(\frac{i-\sqrt{3}}{2}\right)^3$ **(CONCEPTUAL)**